



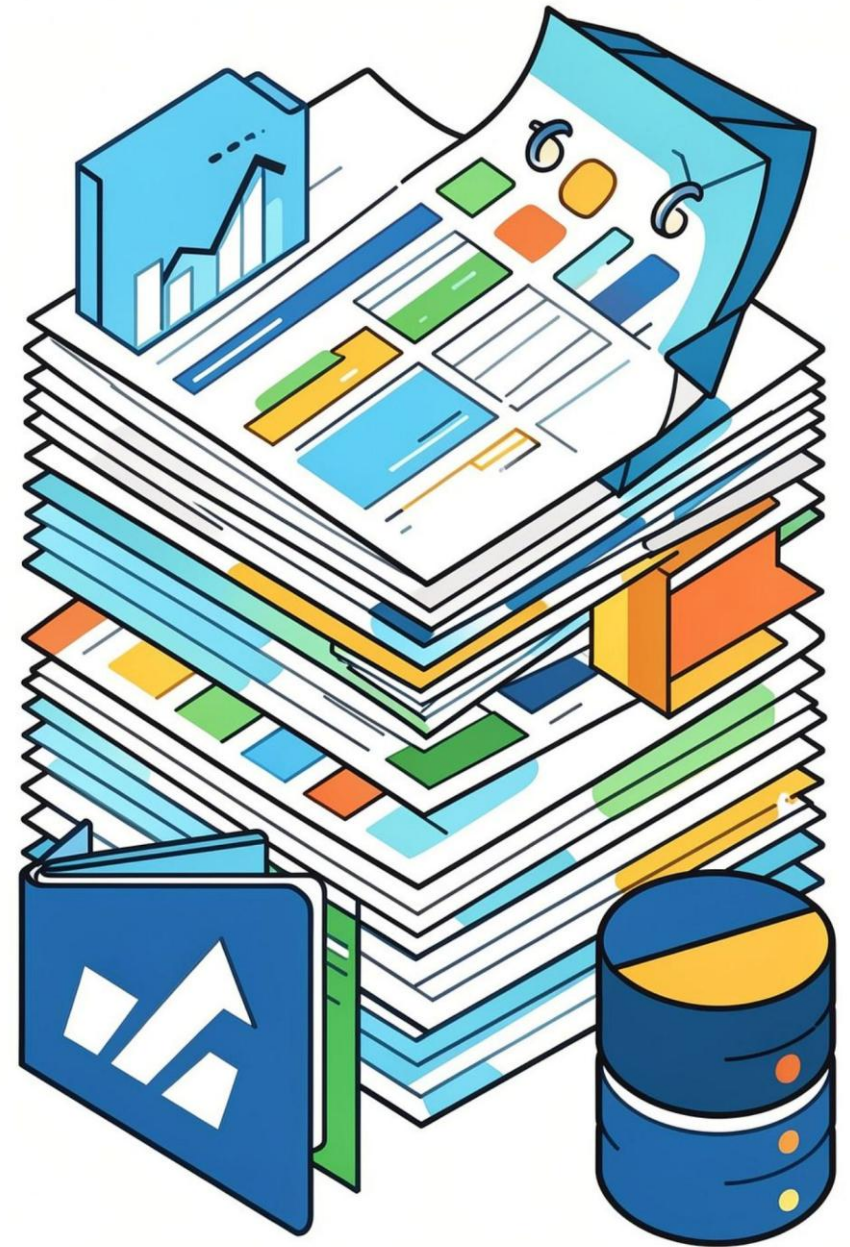
Customer Shopping Behaviour Analysis

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

Project Overview

Objective: extract actionable insights from 3,900 purchases to inform product, pricing, and retention strategy

- Scope: demographics, purchases, shipping, discounts, reviews
- Deliverables: cleaned dataset, SQL analyses, Power BI dashboard, recommendations



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Dataset Snapshot

Rows / Columns

3,900 rows · 18 columns

Key Features

Age, Gender, Location, Item, Category, Amount, Subscription, Shipping, Discounts, Rating

Missing Data

37 missing values in Review Rating (imputed by category median)



Data Prep & Feature Engineering

01

Standardize

Renamed columns to snake_case for clarity

02

Impute

Review rating nulls → category median

03

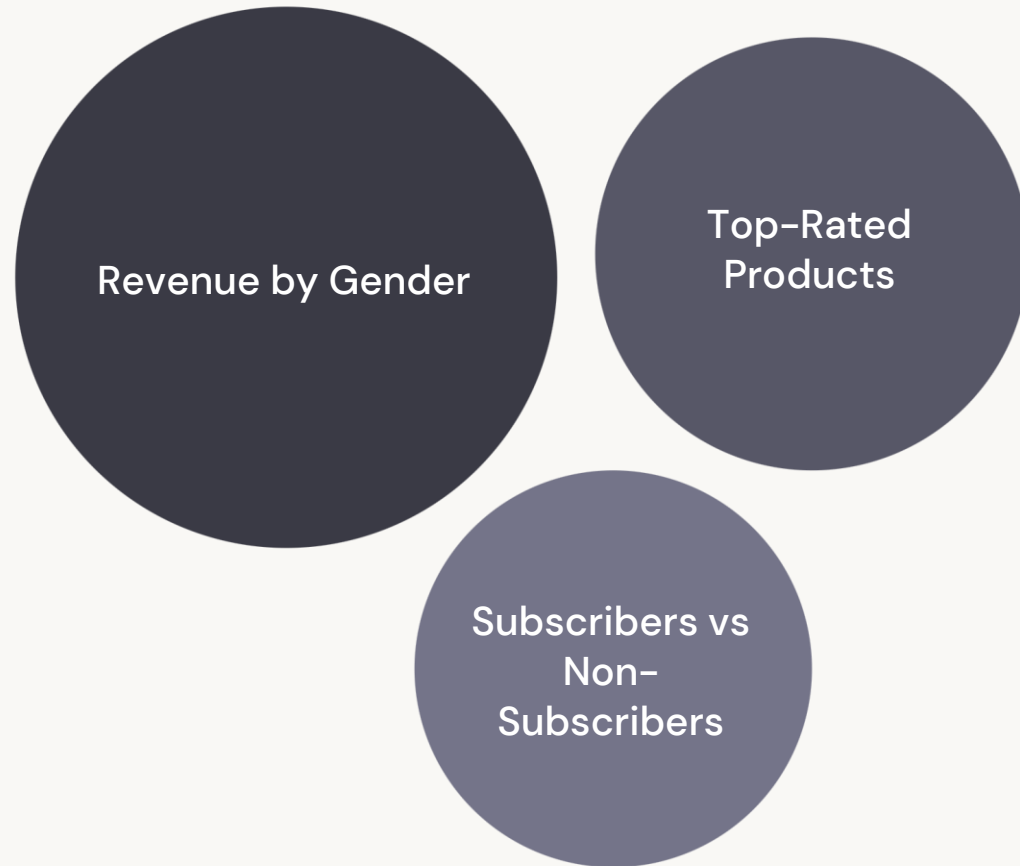
Engineer

age_group bins · purchase_frequency_days · dropped redundant
promo_code_used

04

Load

Exported cleaned table to PostgreSQL for SQL analysis



Key SQL Findings

- Revenue: Male = \$157,890 · Female = \$75,191
- Top-rated items: Gloves, Sandals, Boots (avg ~3.8)
- Subscribers: 1,053 users; avg spend ≈ \$59.49
- Repeat buyers (5+ purchases) skew non-subscribers in counts but higher subscriber conversion

Category Performance (SQL + Dashboard)

Revenue vs Sales volume — use both metrics to prioritize assortment



Clothing

Revenue leader (~100K) ·
Highest sales volume
(≈1,750)



Accessories

Strong revenue (≈75K) ·
steady margin opportunities



Footwear

Lower revenue (~25K) but
top-rated items present

DONUT SUBSCRIPTION PRIT



- Onal

Subscription & Discount Insights

- Subscription rate: 27% subscribed (dashboard)
- Subscribers contribute meaningful recurring revenue — avg spend similar to non-subscribers
- Discount dependence: Hat, Sneakers, Coat, Sweater, Pants have ~47–50% discount rates
- High-spend discount users: 839 rows — opportunity to convert to full-price loyalty

Dashboard Highlights

Top KPIs

Total customers: 3.9K · Avg purchase: \$59.76 · Avg rating: 3.75

Age & Shipping

Revenue by age: Young Adult highest (~62K). Express shipping spends slightly more than Standard.

Product Rank

Category leaders: Blouse, Pants, Shirt;
Accessories: Jewelry, Sunglasses, Belt



Business Recommendations

Boost Subscriptions

Bundle perks → free shipping windows, early access to top-rated items

Reward Loyalty

Target repeat buyers with tiered incentives to reduce discount reliance

Refine Discounting

Limit discounts on high-discount, high-volume SKUs; A/B test margin-preserving offers

Promotion & Merchandising

Highlight top-rated products (Gloves, Sandals) in campaigns to lift conversion

Targeted Acquisition

Focus spend on high-revenue age groups and express-shipping customers